

基

大学院修士課程 理工学研究科

平成 17 年度 筑波大学

入学試験問題

大学院博士課程 システム情報工学研究科
(コンピュータサイエンス専攻)

基礎課題 (構造工学、電子・情報工学 (コンピュータサイエンス)、工学システム分野を志望するものは、
数学を必修とし、物理学または情報基礎のいずれか 1 科目を試験時間中選択。)

情報基礎

Answer both of Problem I and Problem II on separate answer sheets.

Problem I

Answer all questions under the rules of Boolean Algebra. All expressions are logical expressions, and all variables are logical variables if not specially noticed. The symbols “+” and “ $\cdot$ ” represent logical-OR and logical-AND, respectively.

(1) Express the values of the following expressions with $a, \bar{a}$, 0 (false) or 1 (true).

(a) $a + 1$ (b) $a + a$ (c) $a \cdot 0$ (d) $a + \bar{a}$

(2) Prove that the following equation holds. Do not use any truth table.

$$(x + y) \cdot (x + z) = x + y \cdot z$$

(3) Design a logical circuit which compares two unsigned 2-bit binary values A and B through the following steps. The value of A is given as $(a1, a0)$ where $a1$ and $a0$ represent the higher-bit and the lower-bit, respectively. In the same manner, the value of B is given as $(b1, b0)$. The output z gives “1” if A is numerically bigger than B. Otherwise, z gives “0”. Assume that all signals are expressed in positive-logic (active-high). Answer the following questions.

(a) Draw a Karnaugh Map to describe the relationship between the inputs $a1, a0, b1, b0$ and the output z .

Then, describe the output z as a simplified logical expression.

(b) Following the answer of the previous question, draw a logical circuit to output z .

(c) Describe a logical expression of z by the following steps.

The value of A is bigger than that of B if one of the following conditions is satisfied:

(i) The higher-bit of A is bigger than that of B.

(ii) The higher-bits of A and B are the same, and the lower-bit of A is bigger than that of B.

Describe each condition as a logical expression, and express z with them.

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Problem II

A finite set of integers S is represented by an array A in the programming language C as follows:

- $A[0]$ is the number of elements n in S .
- $S = \{A[1], A[2], \dots, A[n]\}$.
- $A[1] < A[2] < \dots < A[n]$.

The next figure shows examples of sets and their representations where “-” is an arbitrary integer.

$\{10\}$	$\iff$	<table><tr><td>1</td><td>10</td><td>—</td><td>—</td><td>—</td><td></td></tr></table>	1	10	—	—	—	
1	10	—	—	—				
$\{30, 10, 20\}$	$\iff$	<table><tr><td>3</td><td>10</td><td>20</td><td>30</td><td>—</td><td></td></tr></table>	3	10	20	30	—	
3	10	20	30	—				
Sets		Arrays						

In the following questions, we assume that A is the array which is declared as follows:

```
#define ASIZE (100)
int A[ASIZE];
```

(1) What is the set represented by the array A after executing the following program ?

```
int i;
A[0] = 10;
for (i=1; i<=10; i++) A[i] = i*i;
```

(2) The function `is_set` returns 1 if the array `A` represents a set in the way described above, and returns 0 otherwise. Complete the following definition of the function `is_set` by filling the box.

```
int is_set () {
    int i;
    if ((A[0] < 0) || (A[0] >= ASIZE))
        return 0;
    for (i=1; i<A[0]; i++) {
        
    }
    return 1;
}
```

(3) Answer the questions (a) and (b) where the function `f` is defined as follows:

```
int f (int val, int i, int j) {
    int k;
    printf("%d %d\n", i, j);
    if (i>j) return 0;
    k = (i+j)/2;    /* "/" is for integer division (which truncates the result). */
    if (A[k] == val) return k;
    else if (A[k] > val)
        return f(val, i, k-1);
    else /* when A[k] < val */
        return f(val, k+1, j);
}
```

In the third line, `printf(...)` is a statement which displays the current values of the variables `i` and `j`.

- (a) Suppose `f(16,1,10)` is executed immediately after the program in the question (1). What are displayed in the course of the execution ?
- (b) Explain what the function `f` returns.